LINUX
COMMAND GRIMOIRE

THE ANIME CHEAT-SHEET EDITION | VOL. 2



- CREATED BY -
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25 CHAPTERS * 378 COMMANDS * 182 NEW MOVES

01 NAVIGATION
Finding your way

`pwd`
print working directory.

`ls`
list directory contents.

`ls -la`
list all files with details.

`cd /path`
change to path.

`cd ..`
go to parent directory.

`cd ~`
go to home directory.

`tree`
display directories as a tree.

`cd -` NEW
jump back to the previous directory.

`ls -lht` NEW
sort by newest, human-readable sizes.

`pushd dir` NEW
save current dir on a stack, then go to dir.

`popd` NEW
return to the last pushed directory.

SENPAI SAYS
Press Tab to auto-complete paths. Fewer typos, more power!

02 FILES & DIRECTORIES
Forge and smash

`touch file.txt`
create an empty file or update timestamp.

`mkdir dir`
create a directory.

`mkdir -p a/b/c`
create nested directories.

`cp src dst`
copy a file or directory.

`cp -r dir1 dir2`
copy a directory recursively.

`mv old new`
move or rename a file or directory.

`rm file`
remove a file.

`rm -rf dir` DANGER
remove a directory tree (use with caution).

`rmdir emptydir`
remove an empty directory.

`stat file`
show file metadata and timestamps.

`basename /path/file`
print filename from path.

`dirname /path/file`
print directory path from filename.

`rm -i file` NEW
ask before deleting (the safe way).

`cp -a src dst` NEW
copy and preserve all attributes.

`ln -s target link` NEW
create a symbolic link.

`file name` NEW
detect what type of file it is.

`realpath file` NEW
print the absolute path.

SENPAI SAYS
Use rm -i when you're nervous. There is no recycle bin here!

03 VIEWING & PAGING
Reading the scrolls

`cat file`
print file contents.

`tac file`
print file contents in reverse.

`head -n 20 file`
show first 20 lines.

`tail -n 20 file`
show last 20 lines.

`tail -f logfile`
follow file updates (logs).

`less file`
page through a file with search.

`more file`
basic pager for files.

`nl file`
number lines while viewing.

`wc -l file`
count lines in a file.

`diff a.txt b.txt` NEW
show differences between two files.

`column -t file` NEW
align output into neat columns.

`zcat file.gz` NEW
read a gzip file without extracting.

`xxd file | head` NEW
peek at raw hex bytes.

SENPAI SAYS
In less, press / to search, n for next match, q to quit.

04 SEARCH & TEXT PROCESSING
Spells of the scribe

`grep "pattern" file`
search pattern in a file.

`grep -ri "pattern" dir/`
recursive, case-insensitive search.

`sort file`
sort lines in a file.

`uniq file`
omit repeated adjacent lines.

`cut -d ':' -f 1 file`
select delimited fields.

`awk '{print $1}' file`
print the first column.

`sed -i 's/old/new/g' file`
in-place find and replace.

`tr 'a-z' 'A-Z'`
translate characters (lower to upper).

`paste file1 file2`
merge lines side by side.

`tee out.txt`
write output to file and stdout.

`grep -rn "text" .` NEW
search with line numbers.

`grep -v "text" file` NEW
show lines that do NOT match.

`grep -c "text" file` NEW
count matching lines.

`sort -n file` NEW
sort numerically.

`uniq -c` NEW
count duplicate lines.

`jq . file.json` NEW
pretty-print and query JSON.

`xargs cmd` NEW
build commands from piped input.

SENPAI SAYS
Chain these with pipes (|) to build your own mini-jutsu... er, tools!

05 FIND & LOCATE
The seeker's guild

`find . -name "*.log"` NEW
find files by name.

`find / -type f -size +100M` NEW
find files bigger than 100 MB.

`find . -mtime -1` NEW
files changed in the last day.

`find . -type d -name src` NEW
find directories by name.

`find . -name "*.tmp" -delete` DANGER NEW
find and delete matching files.

`find . -type f -exec chmod 644 {} +` NEW
run a command on every match.

`locate name` NEW
fast filename search from an index.

`sudo updatedb` NEW
refresh the locate index.

`whereis cmd` NEW
show binary, source and man page paths.

`type cmd` NEW
tell if it is alias, builtin or file.

SENPAI SAYS
find is slow but thorough. locate is fast but needs updatedb first.

06 PERMISSIONS & OWNERSHIP
Seals and barriers

`chmod 644 file`
set rw-r--r-- permissions.

`chmod 755 file`
set rwxr-xr-x permissions.

`chmod +x script.sh`
make a file executable.

`chown user:group file`
change owner and group.

`chgrp group file`
change group ownership.

`umask 022`
set default permissions mask.

`getfacl file`
view file ACLs.

`setfacl -m u:user:rwx file`
set ACL for a user.

`chmod 600 file` NEW
owner-only read/write (keys, secrets).

`chmod -R 755 dir/` NEW
apply permissions recursively.

`chown -R user: dir/` NEW
change owner recursively.

`ls -l` NEW
view permissions at a glance.

SENPAI SAYS
Remember: r=4, w=2, x=1. So 7 = rwx, 6 = rw-, 5 = r-x, 4 = r--.

07 USERS & GROUPS
Guild registry

`whoami` NEW
print the current username.

`id` NEW
show UID, GID and groups.

`who` NEW
list logged-in users.

`w` NEW
who is on and what they run.

`last` NEW
show login history.

`sudo cmd` NEW
run a command as root.

`sudo -i` NEW
open a root login shell.

`su - user` NEW
switch to another user.

`useradd -m name` NEW
create a user with a home dir.

`passwd user` NEW
set or change a password.

`usermod -aG group user` NEW
add a user to a group.

`groups user` NEW
list a user's groups.

`userdel -r user` DANGER NEW
delete a user and their home.

`visudo` NEW
safely edit the sudoers file.

SENPAI SAYS
Prefer sudo for one-off admin tasks instead of logging in as root.

08 PROCESSES & JOBS
Summoning and banishing

`ps aux`
list all processes with details.

`top`
interactive process viewer.

`htop`
enhanced process viewer (if installed).

`pgrep name`
find PIDs by process name.

`pkill name`
send signals by process name.

`kill PID`
terminate a process by PID.

`killall name`
terminate processes by name.

`nice -n 10 cmd`
start with adjusted priority.

`renice -n 10 -p PID`
change priority of running process.

`jobs`
list shell jobs.

`bg %1`
resume job in background.

`fg %1`
bring job to foreground.

`nohup command &`
ignore hangups and run in background.

`time command`
measure execution time.

`watch -n 1 free`
rerun a command at intervals.

`kill -9 PID` DANGER NEW
force-kill a stuck process.

`pstree` NEW
show processes as a tree.

`lsof -i :80` NEW
what is using port 80?

`tmux new -s work` NEW
start a persistent terminal session.

`tmux attach -t work` NEW
reattach to that session.

`Ctrl+C / Ctrl+Z` NEW
interrupt / suspend current job.

SENPAI SAYS
Try kill PID first (polite). Use kill -9 only if the process ignores you.

09 SYSTEM INFO & LOGS
Reading the world map

`uname -a`
kernel and system information.

`hostnamectl`
view or set hostname.

`uptime`
show load averages and uptime.

`free -h`
memory and swap usage.

`vmstat`
virtual memory statistics.

`iostat`
CPU and I/O statistics.

`mpstat`
per-CPU usage statistics.

`pidstat`
per-process statistics.

`lscpu`
CPU architecture details.

`lsusb`
list USB devices.

`lspci`
list PCI devices.

`lshw`
hardware inventory.

`dmesg`
kernel ring buffer messages.

`journalctl -xe`
view recent critical system logs.

`journalctl -u nginx`
view logs for a systemd unit.

`cat /etc/os-release` NEW
which distro and version?

`journalctl -f` NEW
follow system logs live.

`journalctl --since today` NEW
only today's log entries.

`timedatectl` NEW
show or set time and timezone.

`systemd-analyze blame` NEW
what slowed down boot?

`lsmod` NEW
list loaded kernel modules.

SENPAI SAYS
journalctl -f is like tail -f for the whole system. Handy!

10 DISKS & FILESYSTEMS

Treasure vaults

`df -h`
mounted filesystem disk usage.

`du -sh dir/`
directory size summary.

`lsblk`
list block devices.

`blkid`
print block device attributes.

`fdisk -l`
list partition tables.

`parted -l`
list partitions with parted.

`mkfs.ext4 /dev/sdX1` DANGER
create an ext4 filesystem.

`fsck /dev/sdX1` DANGER
check and repair filesystem.

`mount /dev/sdX1 /mnt`
mount a partition.

`umount /mnt`
unmount a filesystem.

`cat /etc/fstab`
view persistent mount table.

`du -h --max-depth=1` NEW
size of each subfolder.

`findmnt` NEW
show mounts as a tree.

`swapon --show` NEW
list active swap areas.

`smartctl -a /dev/sda` NEW
disk health report.

`ncdu` NEW
interactive disk usage browser.

SENPAI SAYS
Always double-check the device name (sdX) before mkfs. Wrong disk = tears.

11 NETWORKING & DNS

Gates between worlds

`ip addr`
show IP addresses.

`ip route`
show routing table.

`ping -c 4 host`
test connectivity.

`traceroute host`
trace network path.

`ss -tuln`
list listening sockets and ports.

`netstat -tuln`
legacy sockets and ports list.

`nslookup example.com`
DNS query.

`dig example.com`
detailed DNS query.

`curl -I https://example.com`
HTTP request.

`wget https://example.com/file`
download a file.

`ssh user@host`
connect via SSH.

`scp file user@host:/path/`
copy file over SSH.

`sftp user@host`
interactive file transfer over SSH.

`nc -l -p 1234`
listen on TCP port.

`nc host 80`
connect to TCP port.

`tcpdump -i eth0 port 443`
capture packets.

`nmap -sS host`
scan ports with SYN.

`ufw enable`
enable uncomplicated firewall.

`ufw allow 22`
allow SSH.

`iptables -L`
list firewall rules.

`firewall-cmd --list-all`
show firewalld config.

`ip -br a` NEW
brief, tidy address list.

`ss -tulpn` NEW
listening ports plus owning process.

`dig +short example.com` NEW
just the answer, nothing else.

`curl -O url` NEW
download keeping the filename.

`curl -s url | jq .` NEW
fetch an API and format JSON.

`mtr host` NEW
live traceroute + ping combined.

`hostname -I` NEW
show all local IP addresses.

`ufw status verbose` NEW
show firewall state and rules.

`nmcli device status` NEW
NetworkManager device overview.

SENPAI SAYS
Ping works but the site doesn't? Check DNS next with dig.

12 SSH & SECURITY

Shield and sword

`ssh-keygen -t ed25519` NEW
generate a modern SSH key pair.

`ssh-copy-id user@host` NEW
install your key on a server.

`ssh -i key.pem user@host` NEW
connect with a specific key.

`ssh -L 8080:localhost:80 host` NEW
forward a local port via SSH.

`sha256sum file` NEW
compute a checksum.

`sha256sum -c sums.txt` NEW
verify checksums.

`gpg --encrypt -r id file` NEW
encrypt a file for a recipient.

`openssl rand -hex 16` NEW
generate a random hex secret.

`fail2ban-client status` NEW
show banned-IP jails.

`sudo ufw status` NEW
quick firewall check.

SENPAI SAYS
Use keys, not passwords. ed25519 keys are short, fast and strong.

13 ARCHIVE & SYNC

Pocket dimension

`tar -czf archive.tar.gz dir/`
create gzip tarball.

`tar -xzf archive.tar.gz`
extract gzip tarball.

`tar -cf archive.tar dir/`
create tar archive.

`tar -xf archive.tar`
extract tar archive.

`zip -r archive.zip dir/`
create zip archive.

`unzip archive.zip`
extract zip archive.

`gzip file`
compress with gzip.

`gunzip file.gz`
decompress gzip file.

`bzip2 file`
compress with bzip2.

`bunzip2 file.bz2`
decompress bzip2 file.

`xz file`
compress with xz.

`unxz file.xz`
decompress xz file.

`rsync -avz src/ dest/`
sync files locally.

`rsync -avz -e ssh src/ user@host:/dest/`
sync over SSH.

`cpio -ov < filelist > archive.cpio`
create cpio archive.

`tar -tf archive.tar` NEW
list contents without extracting.

`tar -xzf a.tar.gz -C /dir` NEW
extract into a chosen folder.

`unzip -l archive.zip` NEW
list zip contents.

`rsync -avzn src/ dest/` NEW
dry run: preview only.

`rsync -avz --delete src/ dest/` DANGER NEW
mirror and delete extras.

`split -b 100M file part_` NEW
cut a big file into chunks.

`dd if=in of=out bs=4M` DANGER NEW
raw copy (double-check of=!).

SENPAI SAYS
Mnemonic: c=create, x=extract, z=gzip, f=file. So tar -czf!

14 PACKAGES: DEBIAN / UBUNTU

Item shop (apt realm)

`apt-get update`
refresh package lists.

`apt-get install nginx`
install a package.

`apt-get upgrade`
upgrade installed packages.

`apt-get remove pkg`
remove a package.

`apt-cache search nginx`
search packages.

`apt-cache show pkg`
show package details.

`dpkg -i pkg.deb`
install a .deb file.

`dpkg -r pkg`
remove a .deb package.

`apt autoremove` NEW
clean up unused dependencies.

`apt list --installed` NEW
list installed packages.

`dpkg -l | grep name` NEW
is a package installed?

`dpkg -L pkg` NEW
list files a package installed.

SENPAI SAYS
Run apt-get update before install so you get fresh package lists.

15 PACKAGES: RHEL & MORE

Item shop (rpm realm)

`yum install pkg`
install a package.

`yum update`
update packages.

`yum remove pkg`
remove a package.

`dnf install pkg`
install with dnf.

`dnf update`
update with dnf.

`rpm -i pkg.rpm`
install an RPM.

`rpm -e pkg`
erase an RPM.

`dnf search name` NEW
search for packages.

`dnf history` NEW
review past transactions.

`rpm -qa` NEW
list all installed RPMs.

`pacman -Syu` NEW
Arch: full system upgrade.

`snap install pkg` NEW
install a snap package.

`flatpak install app` NEW
install a Flatpak app.

SENPAI SAYS
dnf is the modern yum. On new systems, reach for dnf first.

16 SERVICES & SCHEDULING

Familiars and timers

`systemctl start nginx`
start a service.

`systemctl stop nginx`
stop a service.

`systemctl restart nginx`
restart a service.

`systemctl status nginx`
check service status.

`systemctl enable nginx`
enable at boot.

`systemctl disable nginx`
disable at boot.

`service nginx start`
legacy start command.

`service nginx status`
legacy status command.

`crontab -e`
edit cron jobs.

`crontab -l`
list cron jobs.

`crontab -r` DANGER
remove all cron jobs.

`at 09:00`
schedule one-time job.

`batch`
run when load is low.

`sleep 5s`
delay for five seconds.

`systemctl enable --now nginx` NEW
enable at boot and start now.

`systemctl daemon-reload` NEW
reload unit files after edits.

`systemctl --failed` NEW
list units that failed.

`systemctl list-timers` NEW
show scheduled systemd timers.

`atq` NEW
list pending at jobs.

SENPAI SAYS
*Cron time order: minute hour day month weekday. E.g. */5 * * * * runs every 5 min.*

17 ENVIRONMENT & SHELL

Your character sheet

`env`
print environment variables.

`echo $VAR`
print a variable value.

`export VAR=value`
set an environment variable.

`which command`
show command path.

`history`
show command history.

`alias ll='ls -la'`
create a command alias.

`unalias ll`
remove an alias.

`echo $PATH` NEW
show where commands are searched.

`source ~/.bashrc` NEW
reload your shell config.

`man cmd` NEW
open the manual page.

`cmd --help` NEW
quick usage summary.

`Ctrl+R` NEW
reverse-search your history.

`Ctrl+A / Ctrl+E` NEW
jump to line start / end.

`Ctrl+L` NEW
clear the screen.

`!!` NEW
repeat the last command.

SENPAI SAYS
!! repeats the last command. Try sudo !! after a 'permission denied'.

18 PIPES & REDIRECTION
Combining techniques

`cmd > file` NEW
write output to file (overwrite).

`cmd >> file` NEW
append output to file.

`cmd 2> err.txt` NEW
send errors to a file.

`cmd &> all.txt` NEW
send output and errors together.

`cmd < file` NEW
feed a file into a command.

`cmd1 | cmd2` NEW
pipe output into another command.

`cmd1 && cmd2` NEW
run cmd2 only if cmd1 succeeds.

`cmd1 || cmd2` NEW
run cmd2 only if cmd1 fails.

`echo $(date)` NEW
command substitution.

`cmd > /dev/null` NEW
throw away output.

SENPAI SAYS
Pipes pass output as input. Redirects send output to files. Master both!

19 GIT BASICS
Time-travel magic

`git init` NEW
start a new repository.

`git clone url` NEW
copy a remote repository.

`git status` NEW
what changed?

`git add .` NEW
stage all changes.

`git commit -m "msg"` NEW
save a snapshot.

`git push` NEW
upload commits.

`git pull` NEW
download and merge updates.

`git log --oneline --graph` NEW
compact history graph.

`git switch -c name` NEW
create and move to a new branch.

`git diff` NEW
show unstaged changes.

`git stash` NEW
shelve changes temporarily.

`git restore file` NEW
discard changes to a file.

`git reset --hard` DANGER NEW
throw away ALL local changes.

SENPAI SAYS
Commit small and often. Future you will say thanks!

20 VIM & NANO SURVIVAL
Escape the dungeon

`vim file` NEW
open a file in Vim.

`i` NEW
enter insert mode (start typing).

`Esc` NEW
back to normal mode.

`:w :q :wq` NEW
save / quit / save and quit.

`:q!` NEW
quit without saving.

`/text` NEW
search forward (n = next).

`dd yy p` NEW
delete line / copy line / paste.

`u` NEW
undo.

`nano file` NEW
open a file in Nano (beginner-friendly).

`Ctrl+O, Ctrl+X` NEW
nano: save, then exit.

SENPAI SAYS
Stuck in Vim? Press Esc, then type :q! and Enter. You're free!

21 DOCKER
Container summoning

`docker run image`
run a container.

`docker ps`
list running containers.

`docker ps -a`
list all containers.

`docker exec -it container bash`
open a shell in a container.

`docker logs container`
show container logs.

`docker build -t image .`
build an image.

`docker rmi image`
remove an image.

`docker-compose up`
start multi-container app.

`docker-compose down`
stop and remove app.

`docker images` NEW
list local images.

`docker pull image` NEW
download an image.

`docker stop container` NEW
stop a running container.

`docker rm container` NEW
delete a stopped container.

`docker run -d -p 8080:80 nginx` NEW
run detached with a port mapping.

`docker run --rm -it ubuntu bash` NEW
throwaway interactive container.

`docker stats` NEW
live CPU and memory per container.

`docker compose up -d` NEW
start the stack in the background.

`docker system prune` DANGER NEW
remove unused data.

SENPAI SAYS
Add --rm to docker run for throwaway containers that clean up after themselves.

22 KUBERNETES & HELM
The grand fleet

`kubectl get pods`
list pods.

`kubectl get nodes`
list nodes.

`kubectl get services`
list services.

`kubectl apply -f file.yaml`
apply configuration.

`kubectl logs pod`
view pod logs.

`kubectl exec -it pod -- bash`
exec into a pod.

`kubectl describe pod podname`
describe a pod.

`kubectl scale deployment name --replicas=3`
scale a deployment.

`kubectl rollout restart deployment name`
restart a deployment.

`helm install release chart`
install a Helm chart.

`helm upgrade release chart`
upgrade a release.

`helm list`
list releases.

`helm delete release`
delete a release.

`kubectl get pods -A` NEW
pods in every namespace.

`kubectl logs -f pod` NEW
follow pod logs live.

`kubectl port-forward pod 8080:80` NEW
reach a pod from localhost.

`kubectl top pods` NEW
CPU and memory per pod.

`kubectl config get-contexts` NEW
list clusters you can talk to.

`kubectl delete pod name` DANGER NEW
delete a pod.

`helm repo add name url` NEW
add a chart repository.

SENPAI SAYS
Add -n namespace (or -A for all namespaces) when pods seem to be missing.

23 CLOUD CLIS
Skyward kingdoms

`aws configure`
set up AWS CLI credentials.

`aws s3 cp file s3://bucket/`
upload a file to S3.

`aws ec2 describe-instances`
list EC2 instances.

`az login`
authenticate Azure CLI.

`az vm list`
list Azure VMs.

`gcloud auth login`
authenticate gcloud.

`gcloud compute instances list`
list GCP instances.

`aws s3 ls` NEW
list your S3 buckets.

`aws s3 sync dir s3://bucket/` NEW
sync a folder to S3.

`aws sts get-caller-identity` NEW
who am I in AWS?

`az account show` NEW
current Azure subscription.

`gcloud config list` NEW
current gcloud project and account.

SENPAI SAYS
Run aws sts get-caller-identity first. Know which account you're in!

24 SHUTDOWN & REBOOT
The final curtain

`shutdown -h now` DANGER
power off immediately.

`shutdown -r now` DANGER
reboot immediately.

`shutdown -h +10` DANGER
schedule power off in 10 minutes.

`reboot` DANGER
restart the system.

`poweroff` DANGER
power off the system.

`halt` DANGER
halt the system.

`init 0` DANGER
switch to shutdown runlevel.

`init 6` DANGER
switch to reboot runlevel.

`shutdown -c` NEW
cancel a scheduled shutdown.

`systemctl suspend` NEW
sleep, keep state in RAM.

`last -x shutdown` NEW
review past shutdowns.

SENPAI SAYS
Scheduled a shutdown by mistake? shutdown -c cancels it.

25 BOSS BATTLE: COMBO MOVES

One-liners that win fights

`du -h --max-depth=1 | sort -rh | head` NEW
find the biggest folders.

`ps aux --sort=-%mem | head` NEW
top memory hogs.

`ss -tulpn | grep :80` NEW
who owns port 80?

`grep -rn "TODO" . | wc -l` NEW
count TODOs in a project.

`history | awk '{print $2}' | sort | uniq -c | sort -rn | head` NEW
your most used commands.

`tar -czf backup-$(date +%F).tar.gz dir/` NEW
dated backup archive.

`journalctl -u nginx --since "1 hour ago"` NEW
recent logs for a service.

`watch -n 2 'df -h'` NEW
monitor disk space live.

`for f in *.txt; do mv "$f" "${f%.txt}.md"; done` NEW
bulk rename .txt to .md.

`docker ps -q | xargs docker stop` DANGER NEW
stop all running containers.

`find . -name "*.log" -mtime +7 -delete` DANGER NEW
delete logs older than a week.

SENPAI SAYS
Combine small commands into big attacks. This is the Linux way!

